

Part I: Multiple Choice

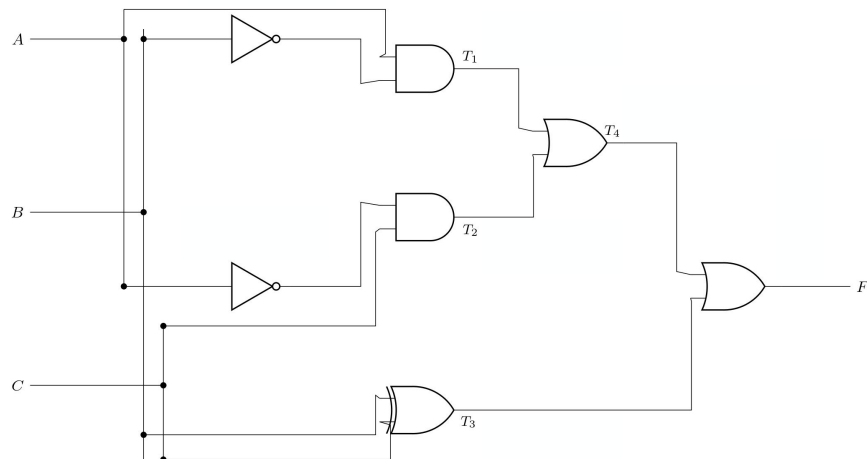
Fill in the best answer. There is only one correct choice for each question.

1. Which of the following best distinguishes a *combinational* circuit from a *sequential* one? [2 pts]
 - ☐ A. A combinational circuit may contain memory elements, while a sequential one may not.
 - ☐ B. A combinational circuit's outputs depend only on the present inputs, while a sequential circuit's outputs also depend on stored internal state.
 - ☐ C. A combinational circuit can only use NAND gates, while a sequential circuit can use any gate.
 - ☐ D. A combinational circuit always has exactly one output, while a sequential circuit has many.
2. In a full adder, what are the three inputs it adds together? [2 pts]
 - ☐ A. Two data bits and a carry-in from the previous, lower position.
 - ☐ B. Two data bits and the final carry-out.
 - ☐ C. A sum bit, a carry bit, and a mode bit.
 - ☐ D. Three data bits with no carry.
3. In a ripple-carry adder, why does a longer chain of full adders take more time to produce a correct result? [2 pts]
 - ☐ A. Each additional adder needs its own clock, and the clocks must synchronize.
 - ☐ B. The carry must propagate through each stage in turn, so the final carry cannot settle until every earlier carry has settled.
 - ☐ C. Longer chains require larger voltages, which take longer to reach.
 - ☐ D. The sum bits must be computed twice for accuracy.
4. How can a decoder be used to implement an arbitrary Boolean function? [2 pts]
 - ☐ A. By wiring the function's inputs to the decoder's outputs.
 - ☐ B. By OR-ing together the decoder output lines that correspond to the function's minterms.
 - ☐ C. By connecting every decoder output to a single AND gate.
 - ☐ D. By using the decoder's enable input as the function's output.
5. In a BCD (decimal) adder, when must the correction value 0110 be added to the raw 4-bit binary sum? [2 pts]
 - ☐ A. Only when both input digits are greater than 5.
 - ☐ B. On every addition, regardless of the sum.
 - ☐ C. Whenever the raw sum exceeds 9 (i.e. it produces a carry, or lands on an illegal code 1010–1111).
 - ☐ D. Only when the carry-in to the digit is 1.

Part II: Show Work

Show all work for full credit. Partial credit will be given.

1. **Circuit analysis.** The combinational circuit below has three inputs A , B , C and a single output F . The intermediate gates have been labeled T_1 , T_2 , T_3 , and T_4 .

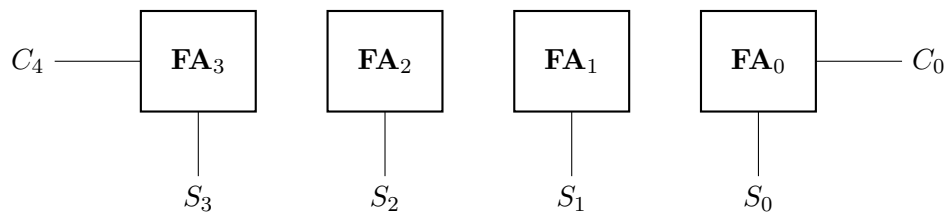


- (a) First write the Boolean function for each gate that depends only on the input variables (T_1 , T_2 , T_3). Then write the functions for the gates that depend on already-labeled gates (T_4 and F). Finally, by repeated substitution, express F purely in terms of A , B , and C . [6 pts]

- (b) Complete the truth table below, filling in each intermediate gate column (T_1 , T_2 , T_3 , T_4) before combining them to get F . [8 pts]

A	B	C	T_1	T_2	T_3	T_4	F
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					

2. **Ripple subtractor.** The chain of full adders below is wired as a 4-bit subtractor and is used to compute $A - B$ with $A = 1010$ and $B = 0110$. Only the full-adder blocks, the sum output lines, the carry-in line C_0 , and the final carry line C_4 are drawn; you must supply everything else. [14 pts]



- (a) For a subtraction, what value is placed on C_0 , and what does each XOR gate do to the bits of B before they enter the adders? [3 pts]
- (b) On the diagram above, write the value entering each full adder for A_i and for B_i (after the XOR), and label the internal carries C_1, C_2, C_3 . [4 pts]
- (c) Fill in each sum output $S_3S_2S_1S_0$ on the diagram, then write the 4-bit result here: [4 pts]
- (d) Read the final carry C_4 off your diagram and explain, in one sentence, what its value tells you about A and B in this subtraction. [3 pts]

3. **Design procedure.** Design a combinational circuit with three inputs x, y, z and two outputs F_1 and F_2 , defined as follows:

- $F_1 = 1$ exactly when the input, read as the binary number xyz (with x most significant), is 5 or greater.
- $F_2 = 1$ exactly when the input contains an *odd* number of 1's.

(a) Fill in the truth table for both outputs. [5 pts]

x	y	z	F_1	F_2
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

(b) Write each output as a sum of minterms and then reduce it to its simplest form. [5 pts]

(c) Sketch a logic diagram that realizes both outputs using AND, OR, and NOT gates; label the inputs and mark F_1 and F_2 . [4 pts]

4. **BCD adder.** Using the add-in-binary-then-correct method from class, add the BCD digits $9+8$.

(a) Add the two 4-bit codes in a binary adder and record the raw sum $K Z_8 Z_4 Z_2 Z_1$. [4 pts]

(b) Using $C = K + Z_8 Z_4 + Z_8 Z_2$, decide whether the raw sum needs correcting, and say why. [4 pts]

(c) Apply the correction if needed and give the final answer as a carry digit plus a valid BCD digit; confirm it equals the expected decimal value. [4 pts]

5. **Decoder.** You are given a 3-to-8-line decoder (active-high outputs D_0 through D_7) and OR gates. Use them to build $F(x, y, z) = \sum m(0, 2, 3, 5, 7)$.

(a) In one or two sentences, explain why a decoder plus an OR gate can realize any function expressed as a sum of minterms. [3 pts]

(b) Which decoder output lines must be OR'd together to form F ? [4 pts]

(c) Draw the decoder with x, y, z on its inputs and the OR gate wired to produce F . [3 pts]

6. **Multiplexer.** Implement $F(A, B, C) = \sum m(0, 3, 4, 6, 7)$ with a single 8-to-1 multiplexer whose selection lines $S_2S_1S_0$ are driven by A, B, C (A most significant).

(a) Write the truth table for F .

[4 pts]

A	B	C	F
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

(b) State the constant (0 or 1) tied to each data input I_0 through I_7 .

[4 pts]

(c) Draw the multiplexer with its data inputs wired to those constants and A, B, C on the selection lines.

[4 pts]

7. **Adder / subtractor.** Consider the adder-subtractor circuit studied in class (four full adders, XOR gates on each B input driven by mode line M , end carry C_4 , and overflow output V). Calculate the result of the given binary arithmetic equation and label the coefficients correctly. [14 pts]

$$A = 0111, \quad B = 0010, \quad M = 1.$$

- (a) What are the values of A_0 to A_3 ? [2 pts]

$$A_0 = \boxed{} \quad A_1 = \boxed{} \quad A_2 = \boxed{} \quad A_3 = \boxed{}$$

- (b) What are the values of B_0 to B_3 that actually enter the adders (after the XOR gates)? [2 pts]

$$B_0 = \boxed{} \quad B_1 = \boxed{} \quad B_2 = \boxed{} \quad B_3 = \boxed{}$$

- (c) What are the values of C_0 to C_4 ? [3 pts]

$$C_0 = \boxed{} \quad C_1 = \boxed{} \quad C_2 = \boxed{} \quad C_3 = \boxed{} \quad C_4 = \boxed{}$$

- (d) What are the values of S_0 to S_3 ? [3 pts]

$$S_0 = \boxed{} \quad S_1 = \boxed{} \quad S_2 = \boxed{} \quad S_3 = \boxed{}$$

- (e) What is the value of V ? [2 pts]

- (f) Does this operation result in an overflow? [2 pts]

8. **Extra Credit.** If you could instantly master any one skill or field of knowledge, what would it be, and what is the first thing you would do with it? **[3 pts]**